

Multiple Choice

1. **Answer:** E

Options: A) - 10 mph; B) 25 mph; C) -30 mph; D) -15 mph; E) - 5 mph

Note: Correct option: E - - 5 mph

2. **Answer:** A

Options: A) 100; B) 75; C) 80; D) 70; E) 60 *Note:* Correct option: A - 100

2. **Answer:** A

Options: A) 3/4; B) 2/3; C) 1/4; D) 5/8; E) 7/8

Note: Correct option: A - 3/4

3. **Answer:** C

Options: A) FIFO memory; B) Flash memory; C) LIFO memory; D) LILO memory

Note: Correct option: C - LIFO memory

4. **Answer:** E

Options: A) $P(R) = 2/3$, $P(B) = 1/3$; B) $P(R) = 5/14$, $P(B) = 4/14$; C) $P(R) = 1/2$, $P(B) = 1/4$; D) $P(R) = 7/15$, $P(B) = 4/15$; E) $P(R) = 7/16$, $P(B) = 1/4$

Note: Correct option: E - $P(R) = 7/16$, $P(B) = 1/4$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' p^2 / (mRT)'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '45,000 lb'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** $y[n-1] = y[n] + n^2$; $n \geq 1$ where $y[0] = 0$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 3.6 newtons. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.